Augmented Reality Circuit Tracing

Community Partner: Schweitzer Engineering Laboratories
Contact Name: Cody Tews
Contact email: cody_tews@selinc.com
Contact phone: 509-592-0461

IP and Compliance Requirements

- Students must sign an NDA: **No**
- Intellectual Property will be owned by: **Sponsor**
- Students must be US persons: **No**
- Other legal requirements (background checks, drug testing, etc): **None**

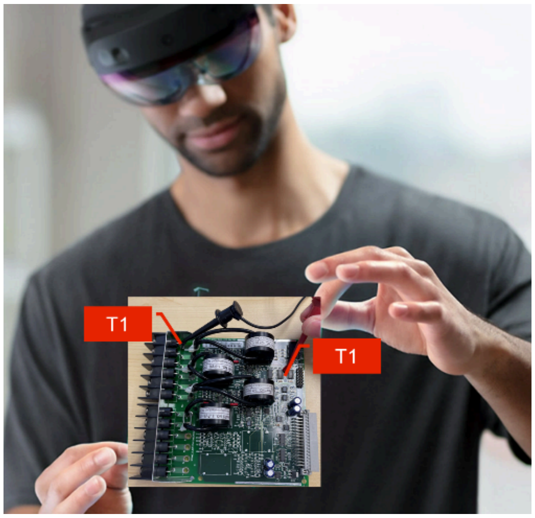
Background and Motivation

Schweitzer Engineering Laboratories (SEL) is a large, US-based company which invents and manufactures equipment which measures, controls, and protects the electric power system. Once installed the equipment will have many cable connections to provide power, analog sensor inputs, digital sensor inputs, digital outputs, and inter-device communications. Assuring that a wire is attached to the correct terminal can be a labor intensive process. The goal of this project is to use augmented reality technology to reduce the cognitive load on the manufacturing team and produce automated artifacts for the build process.

To learn more about SEL, please visit our website at https://selinc.com/

Objective

Manual circuit tracing is labor intensive as a user must mentally change context between design drawing and the implemented circuit. Additional, labor burden is introduced when the user must also annotate wiring diagrams with notes confirming wiring state.



The use of augmented reality technology may reduce the mental context switching for the user and thus increase overall productivity. Figure 1 provides a mockup of the Augmented Reality Circuit Tracing (ARCT) approach. Digital cameras follow to position of the user's electrical probes. When electrical conductivity is detected, the analysis workstation annotates the image with a common label for the probe positions. In the example image, electrical continuity is detected between the probes, and the user's augmented reality image now includes the labels of T1.

Requirements

Digital cameras may be placed to digitally monitor the probe points. Multiple cameras may be used to capture physically separate probe points. The digital camera feeds may be networked to an analysis computer.

Computer vision software (e.g. OpenCV) will analyze digital camera feeds and perform object detection of measurement probes. The software will extract relative position of test probes.

The test probes will be connected through to an analog-to-digital converter (ADC) that is attached (wired or wirelessly) to the analysis computer (e.g. LabJack). When a positive circuit connection is made, the ADC will receive voltage exceeding a threshold value. This will in turn generate a software message of connectivity.

When a connectivity message is received, the computer vision software will insert an augmented reality label of the new connectivity. Additionally, a log will be generated which captures the relative position of all probes for each positive instance of connectivity.

To provide the user with the augmented reality information several technical approaches are possible. To maximize the application in an offline approach, a rudimentary system can be constructed OpenCV. This library provides capabilities for object detection and image augmentation.

A more immersive experience can be produced by leveraging a dedicated augmented reality platform such as Microsoft's HoloLens platform. This platform provides object and spatial anchoring. Object anchoring allows for the alignment of 3D content to physical objects. Spatial anchoring enables mixed-reality experiences. These technologies may enable a user to change position relative to the device under analysis while still maintaining the relative annotation of circuit connectivity.

Please do not use cloud computing resources (e.g. OpenAI) as it will make the prototype unusable within our corporate network.

Suggested Skills

Existing Skills

- Python programming
- Basic circuits

Developing Skills

- Computer vision
- Augmented reality

Deliverables

At the beginning of the project, SEL would like an exploratory report, summarizing potential technical approaches (e.g. hardware and software) and recommendations. This may be part of the project proposal.

Monthly SEL would like a summary status report (~ 1 page) documenting task progress against the original project proposal and execution plan for the subsequent month.

At project completion SEL would like a working prototype to demonstrate approach viability.